

**Tribhuvan University**  
**Institute of Science and Technology**  
**2068**

Bachelor Level/ First Year/ First Semester/ Science  
**Computer Science and Information Technology (CSc. 111)**  
(Digital Logic)

Full Marks: 60  
Pass Marks: 24  
Time: 3 hours.

Candidates are required to give their answers in their own words as far as practicable.  
The figures in the margin indicate full marks.

**Long Questions:**

**Attempt any two questions: (2 × 10=20)**

1. Draw a block diagram truth table and logic circuit of 1\*16 Demultiplexer and explain its working principle.
2. Design a 3 bit synchronous counter and explain it.
3. What is magnitude comparator? Design a logic circuit for 4 bit comparator and explain it.

**Short Questions:**

**Attempt any eight questions: (8 × 5=40)**

4. Design a half subtractor circuit using only NAND gates.
5. Convert the following decimal numbers into Hexadecimal and Octal numbers:
  6. 504
  7. 250
8. Design an encoder using universal gates.
9. What do you mean by D-flip-flop?
10. What is sequential logic? What are the important features?
11. Simplify the Boolean function using K-Maps.
12.  $F = X'yz + X'yz' + Xy'z' + Xy'z$
13. Draw a parallel-parallel-out shift register and explain it.
14. Explain the 4 bit ripple counter.
15. Explain the programmable logic array.
16. Write short notes on :
  - a. Asynchronous counter
  - b. Multiplexers
  - c. State reduction table